PHYS 311 Formula Sheet

Chukwugozie Okeke

Thermodynamics

Thermodynamic Potentials

Internal	$U(S,V)$	-	$dU = TdS - PdV$
Helmholtz	$F(T,V)$	$U - TS$	$dF = -SdT - PdV$
Enthalpy	$H(S,P)$	$U + PV$	$dH = TdS + VdP$
Gibbs	$G(T,P)$	$U + PV - TS$	$dG = -SdT + VdP$

Conservation of Energy

$$dU = dQ - dW = TdS - PdV \quad C_V = \left(\frac{dU}{dT}\right)_V \quad C_P = \left(\frac{\partial H}{\partial T}\right)_P$$

Ideal Gases

$$PV = Nk_BT \qquad U = \frac{3}{2}Nk_BT \qquad C_P = C_V + R$$

$$\Delta S = \frac{3}{2}Nk_B \ln\left(\frac{T_f}{T_i}\right) + Nk_B \ln\left(\frac{V_f}{V_i}\right)$$

Adiabatic $\implies PV^\gamma = \text{const.}$ $\qquad \gamma = C_P/C_V$

Microcanonical Ensemble

$$S = k_B \ln(W) \qquad ln(M!) \approx M\ln(M) - M$$

Two-state Paramagnet

$$\epsilon_{up/dn} = \mp\mu B \qquad x = \frac{n_{up}}{N} \qquad \frac{U}{\mu B} = (n_{dn} - n_{up}) = (1-2x)N$$

$$S = -k_BN(x\ln(x) + (1-x)\ln(1-x)) \qquad \frac{\mu B}{k_BT} = \frac{1}{2}\ln\left(\frac{x}{1-x}\right)$$

$$x = \frac{1}{1 + e^{-\frac{2\mu B}{k_BT}}} \qquad U = -N\mu B \tanh\left(\frac{\mu B}{k_BT}\right)$$

$$C_B = Nk_B\left(\frac{\mu B}{k_BT}\right)^2 \frac{1}{\cosh^2(\mu B/k_BT)}$$

Rubber Band

$$dU = TdS + Fdl \qquad l = (n_+ - n_-)d = (2n_+ - N)d$$

$$x = \frac{l}{Nd} = \left(\frac{2n_+}{N} - 1\right) \qquad F = \frac{k_B}{2d}\ln\left(\frac{1+x}{1-x}\right) \approx \frac{k_BTl}{Nd^2}$$

$$S = -k_BN\left[\left(\frac{1+x}{2}\right)\ln\left(\frac{1+x}{2}\right) + \left(\frac{1-x}{2}\right)\ln\left(\frac{1-x}{2}\right)\right]$$

Canonical Ensemble

$$Z = \sum_j e^{-\beta E_j} = \sum_n g_n e^{-\beta E_n} \qquad F = -k_BT\ln(Z) \qquad \beta = \frac{1}{k_BT}$$

$$p_i = \frac{e^{E_j/k_BT}}{Z} \qquad S = -k_B\sum_i p_i\ln(p_i) = -k_B\langle\ln(p_i)\rangle$$

Useful Identities

$$U = k_BT^2\left(\frac{\partial\ln(Z)}{\partial T}\right) = -\frac{\partial\ln(Z)}{\partial\beta}$$

Equipartition Theorem $\rightarrow \frac{1}{2}k_B$ per d.o.f.

Two-level System

$$Z = e^{\epsilon/k_BT} + e^{-\epsilon/k_BT} \qquad F = -\epsilon - k_BT\ln(1 + e^{-2\epsilon/k_BT})$$

$$S = k_B\ln(1 + e^{-2\epsilon/k_BT}) + \frac{2\epsilon/T}{e^{2\epsilon/k_BT} + 1}$$

$$C_V = \frac{\epsilon^2}{k_BT\cosh^2(\epsilon/k_BT)} \qquad \langle U\rangle = -\epsilon\tanh(\epsilon/k_BT)$$

Single Particle in a Well

$$Z = \sum_{n=1}^{\infty} e^{-\beta E_n} \qquad E_n = \frac{n^2\pi^2\hbar^2}{2mL^2} \qquad \alpha = \frac{\pi^2\hbar^2}{2mL^2}$$

Low Temp. $\xrightarrow{\alpha\beta<1} Z \approx \left(\frac{mL^2}{2\pi\hbar^2}k_BT\right)^{1/2}$ $\qquad U = \frac{1}{2}k_BT$

$$Z \approx \frac{L}{\lambda_D} \qquad \lambda_D = \left(\frac{2\pi\hbar^2}{mk_BT}\right)^{1/2} \qquad PL = k_BT$$

High Temp. $\xrightarrow{\alpha\beta>>1} Z \approx e^{-\alpha\beta}(1 + e^{-3\alpha\beta} + ...)$ $\qquad PL^3 = \frac{\pi^2\hbar^2}{m}$

Diatomic Molecule

$$Z = Z_{trans} \times Z_{rot} \times Z_{vib}$$

Translational Levels

$$Z_{trans} = L^3\left(\frac{mk_BT}{2\pi\hbar^2}\right)^{3/2} \qquad PV = Nk_BT = 2\langle U\rangle/3 \quad C_V = \frac{3k_B}{2}$$

$$F_{trans} = -k_BT\ln(Z_{trans}) = -k_BT\left(\ln(V) + \frac{3}{2}\ln\left(\frac{mk_BT}{2\pi\hbar^2}\right)\right)$$

$$S = k_B\left(\ln(V) + \frac{3}{2}\ln\left(\frac{mk_BT}{2\pi\hbar^2}\right) + \frac{3}{2}\right)$$

Vibrational Levels

$$Z_{vib} = \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}} \qquad \theta_{vib} = \frac{\hbar\omega}{k_B} \qquad C_V = k_B\left(\frac{\theta_V}{T}\right)^2\frac{e^{\theta_V/T}}{(e^{\theta_V/R} - 1)^2}$$

Low Temp. $\implies C_V \approx \left(\frac{\theta_V}{T}\right)^2 e^{-\theta_V/T}$

High Temp. $\implies C_V \approx k_B\left(1 - \frac{1}{12}\left(\frac{\theta_V}{T}\right)^2\right) \approx k_B$

$$\langle U\rangle = \hbar\omega\left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega} - 1}\right) = \hbar\omega\left(\frac{1}{2} + \langle n\rangle\right)$$

Rotational Levels

$$Z_{rot} = \sum_{l=0}^{\infty}(2l+1)e^{-\beta\epsilon_l} \qquad \epsilon_l = \frac{\hbar^2 l(l+1)}{2I} \qquad \theta_{rot} = \frac{\hbar^2}{2Ik_BT}$$

Low Temp. $Z_{rot} \approx 1 + 3e^{-6\theta_{rot}/T}$

High Temp. $Z_{rot} \approx \frac{T}{\theta_{rot}}$ $\qquad U = k_BT$ $\qquad C_V = k_B$

Identical Particles

$$Z_N \approx \frac{(Z_1)^N}{N!}$$

Sackur-Tetrode Equation

$$S = Nk_B\left(\ln\left(\frac{V}{N}\right) + \frac{3}{2}\ln\left(\frac{mk_BT}{2\pi\hbar^2}\right) + \frac{5}{2}\right)$$

Alt. $S = Nk_B\left(\ln\left(\frac{V}{N}\right) + \frac{3}{2}\ln\left(\frac{mU}{3\pi\hbar^2 N}\right) + \frac{5}{2}\right)$

Maxwell Distribution

Density of States

$$D_V(k) = \frac{V}{2\pi^2}k^2 \qquad D_A(k) = \frac{A}{2\pi}k \qquad D_L(k) = \frac{L}{\pi}$$

$$D(\epsilon) = D(k)\left(\frac{\partial k}{\partial \epsilon}\right)\partial\epsilon$$

Maxwellian Speed Distribution

$$f(v) = \frac{4}{\sqrt{\pi}}\frac{v^2}{v_0^3}e^{-(v/v_0)^2} \qquad v_0 = \sqrt{2k_BT/m}$$

$$\langle v\rangle = \sqrt{\frac{8k_BT}{\pi m}} \qquad \langle v^2\rangle = \frac{3k_BT}{m} \qquad \langle E\rangle = \frac{3}{2}k_BT$$

Maxwellian Position Distribution

$$n(\vec{r}) \propto e^{-U(\vec{r})/k_BT}$$

Maxwellian Beam

$$\Phi(v) = \frac{N}{V\sqrt{\pi}}\left(\frac{v}{v_0}\right)^3 e^{-(v/v_0)^2} \qquad \Phi = \frac{1}{A}\frac{dN}{dt} = \frac{N}{V}\sqrt{\frac{k_BT}{2\pi m}}$$

Planck Distrubtion

Blackbody Radiation and Planck Distribution

$$P = A\epsilon\sigma T^4 \qquad \lambda_m T = 0.002899\text{mK} \qquad u(\lambda) = \frac{8\pi hc}{\lambda^5(e^{hc/\lambda k_BT}-1)}$$

$$u(\lambda) = D(\lambda)U(\lambda,T) = \frac{8\pi}{\lambda^4}U(\lambda,T)$$

$$u(\nu) = \frac{8\pi h}{c^3}\frac{\nu^3}{e^{h\nu/k_BT}-1}$$

$$U_{tot} = VaT^4 \qquad S = \frac{4}{3}aVT^3 \qquad a = \frac{8\pi^5 k_B^4}{15h^3c^3} = \frac{4}{c}\sigma$$

$$F = -\frac{1}{3}aVT^4 \qquad \text{Radiation Pressure} \rightarrow P_{rad} = \frac{1}{3}aT^4$$

General Planck Distribution

$$\epsilon_n(\mathbf{k}) = \left(n+\frac{1}{2}\right)\omega(\mathbf{k}) \qquad p_n(\mathbf{k}) = \frac{e^{-n\hbar\omega(\mathbf{k})/k_BT}}{Z(\mathbf{k})}$$

$$Z(\mathbf{k}) = \frac{1}{1-e^{-\hbar\omega(k)/k_BT}} \qquad \bar{n}(\mathbf{k}) = \frac{1}{e^{\hbar\omega(k)/k_BT}-1}$$

$$\bar{U}(\mathbf{k}) = \hbar\omega(\mathbf{k})\bar{n}(\mathbf{k}) \qquad \bar{U} = \sum_k \bar{U}(\mathbf{k}) \rightarrow \int_0^\infty D(k)\bar{U}(k)dk$$

Einstein Solid Model

$$F = N\epsilon + 3Nk_BT\ln\left(1-e^{-\hbar\omega_E/k_BT}\right)$$

$$C_V = 3Nk_B\left(\frac{\hbar\omega_E}{k_BT}\right)^2\frac{e^{\hbar\omega_E/k_BT}}{\left(e^{\hbar\omega_E/k_BT}-1\right)^2}$$

Debye Solid Model

$$D(\omega)d\omega = \frac{V}{2\pi^2}\omega^2\left(\frac{1}{s_L^3}+\frac{2}{s_T^3}\right)d\omega \approx \frac{3V}{2\pi^2\bar{s}^3}\omega^2 d\omega \text{ if } \omega < \omega_D, \text{ else } 0$$

$$\#\text{DOF} = \int_0^{\omega_D} D(\omega)d\omega \rightarrow \omega_{D,V} = \bar{s}(6\pi^2 N/V)^{1/3} \qquad x = \hbar\omega/k_BT$$

$$F = N\epsilon + \frac{3k_BTV}{2\pi^2\bar{s}^3}\int_0^{\omega_D}\omega^2\ln\left(1-e^{-\hbar\omega/k_BT}\right)d\omega \qquad \theta_D = \hbar\omega_D/T$$

Low temp $\rightarrow$ approx. $\ln(1-x) \approx x$ and integrate to infinity. $U \approx \frac{3\pi}{5}\frac{Nk_BT^4}{5\theta_D^3}$. $C_V \approx \frac{12\pi^4}{5}\left(\frac{T}{\theta_D}\right)^3 Nk_B$

High temp $\rightarrow$ approx. $e^{-x} \approx 1-x$. $U \approx 3Nk_BT$

Grand Canonical Ensemble

Variable Particle Systems

$$\mu = -T\left(\frac{\partial S}{\partial N}\right)_{U,V} = \left(\frac{\partial F}{\partial N}\right)_{V,T} = -k_BT\left(\frac{\partial \ln(Z)}{\partial N}\right)_{V,T}$$

Ideal Gas $\implies \mu = \Delta + k_BT\ln\left(\frac{n}{gn_Q}\right)$ $\quad n_Q = \frac{1}{\lambda_D^3} = \left(\frac{mk_BT}{2\pi\hbar^2}\right)^{3/2}$

Variable Particle Systems Equilibrium Condition

$$\sum V_i[R_i] \rightleftharpoons \sum V_j[P_j] \implies \sum V_i\mu_{Ri} - \sum V_j\mu_{Pj} = 0$$

$\rightarrow$ If $\sum V_i\mu_{Ri} > \sum V_j\mu_{Pj}$ then reaction produces more products. Else if less than 0, reaction produces more reactants.

Saha Formula

$$\frac{n_p n_e}{n_H} = \left(\frac{m_H}{m_e m_p}\frac{2\pi\hbar}{k_BT}\right)^{3/2}e^{B_H/k_BT}$$

$$B_H = (m_H - m_p - m_e)c^2 = -13.6eV$$

Grand Canonical Ensemble

$$\Xi = \sum_j e^{-\beta(E_j - \mu N_j)} \qquad \Phi_G = -k_B T \ln(\Xi) = \bar{U} - \mu\bar{N} - TS$$

$$d\bar{U} = -PdV + \mu d\bar{N} + TdS \qquad d\Phi_G = -PdV - \bar{N}d\mu - SdT$$

Indep., disting. $\rightarrow \Xi = \dfrac{1}{1 - e^{\beta\mu} Z_1}$

Classical, indep., indisting. $\rightarrow \Xi = e^{e^{\beta\mu} Z_1}$

Absorption of Atoms on Surface

$$\Xi = 1 + e^{-(\epsilon_i - \mu)/k_B T} \qquad \langle n_i \rangle = p_{filled} = \frac{1}{e^{(\epsilon_i - \mu)/k_B T} + 1}$$

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Fermi Particles

Single Fermi Particle

$$\Xi_{\mathbf{k}} = 1 + e^{-(\epsilon(\mathbf{k}) - \mu)/k_B T} \qquad \Phi_G = -k_B T \ln\left(1 + e^{-(\epsilon(\mathbf{k}) - \mu)/k_B T}\right)$$

$$n(\mathbf{k}) = -\left(\frac{\partial \Phi_G(\mathbf{k})}{\partial \mu}\right)_{U,V} = \frac{1}{e^{(\epsilon(k) - \mu)/k_B T} + 1} \qquad U(\mathbf{k}) = n(\mathbf{k})\epsilon(\mathbf{k})$$

$$S(\mathbf{k}) = -k_B \left(n(\mathbf{k}) \ln[n(\mathbf{k})] + [1 - n(\mathbf{k})] \ln[n(\mathbf{k})]\right)$$

Gaseous Fermi Particles at Absolute Zero

$$N = 2\int_0^{k_F} D(k)dk \implies k_F = (3\pi^2 n)^{1/3} \qquad E_F = \frac{\hbar^2 k_F^2}{2m}$$

$$p_F = \hbar k_F \qquad v_F = \frac{\hbar k_F}{m} \qquad P_F = \frac{2nE_F}{5} \qquad T_F = \frac{E_F}{k_B}$$

$$\bar{U} = 2\sum_{\mathbf{k}} n(\mathbf{k})U(\mathbf{k}) \rightarrow 2\int_0^{k_F} D(k)U(k)dk = \frac{3NE_F}{5}$$

Gaseous Fermi Particles at Finite Temperature

$$U = \frac{3NE_F}{5} + \frac{\pi^2}{4} N \frac{(k_B T)^2}{\epsilon_F} \qquad C_V = \frac{\pi^2 N k_B^2 T}{2\epsilon_F}$$

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Bose Particles

Single Bose Particle

$$\Xi_{\mathbf{k}} = \frac{1}{1 - e^{-(\epsilon(\mathbf{k}) - \mu)/k_B T}} \qquad \Phi_G = k_B T \ln\left(1 + e^{-(\epsilon(\mathbf{k}) - \mu)/k_B T}\right)$$

$$f(\mathbf{k}) = -\left(\frac{\partial \Phi_G(\mathbf{k})}{\partial \mu}\right)_{U,V} = \frac{1}{e^{(\epsilon(\mathbf{k}) - \mu)/k_B T} - 1} \qquad U(\mathbf{k}) = f(\mathbf{k})\epsilon(\mathbf{k})$$

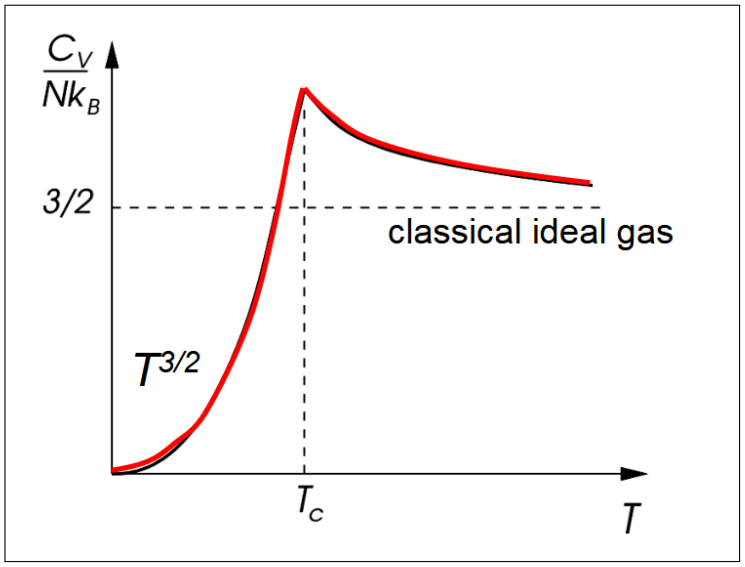
$$S(\mathbf{k}) = k_B \left([1 + f(\mathbf{k})] \ln[1 + f(\mathbf{k})] - [f(\mathbf{k})] \ln[f(\mathbf{k})]\right)$$

Systems of Bose Particle

$$T_c = 3.31 \frac{\hbar^2}{m k_B} \left(\frac{N}{V}\right)^{2/3} \qquad N_0 = N\left(1 - \left(\frac{T}{T_c}\right)^{3/2}\right)$$

For $T \leq T_c$, $\mu = 0$

$$U = 0.770 \left(\frac{T}{T_c}\right)^{3/2} N k_B T \qquad S = 1.284 \left(\frac{T}{T_c}\right)^{3/2} N k_B$$



White Dwarfs and Chandrasekhar Limit

Relativistic Gases $\rightarrow P = \dfrac{\hbar c}{12\pi^2} k_F^4 = \dfrac{(3\pi^2)^{1/3}}{4} \hbar c n^{4/3}$

Gravitational Pressure $P_G \approx \dfrac{GM^2}{R^4}$

Chandrasekhar Limit $M = \sqrt{\dfrac{\hbar^3 c^3}{m_p^4 G^3}} \approx 1.85 M_\odot$. True value is $1.45 M_\odot$, where $M_\odot \approx 1.989 \cdot 10^{30}$kg is one solar mass.

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Useful Identities

$$\bar{U} = -\left(\frac{\partial \ln(Z_1)}{\partial \beta}\right) \qquad \Delta U^2 = \frac{\partial^2 \ln(Z_1)}{\partial \beta^2}$$

$$\Delta N^2 = k_B T \left(\frac{\partial \bar{N}}{\partial \mu}\right) = (k_B T)^2 \frac{\partial^2 \ln(\Xi)}{\partial \mu^2}$$

Physical Constants

$\sigma = 5.670 \cdot 10^{-8}$ W/m^{2}K^4 $\qquad a = (4/c)\sigma$

Trigonometry:

$$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$$
$$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$$

Law of cosines:

$$c^2 = a^2 + b^2 - 2ab\cos\theta$$

Integrals:

$$\int x \sin(ax)\, dx = \frac{1}{a^2}\sin(ax) - \frac{x}{a}\cos(ax)$$
$$\int x \cos(ax)\, dx = \frac{1}{a^2}\cos(ax) + \frac{x}{a}\sin(ax)$$

Exponential integrals:

$$\int_0^\infty x^n e^{-x/a}\, dx = n!\, a^{n+1}$$

Gaussian integrals:

$$\int_0^\infty x^{2n} e^{-x^2/a^2}\, dx = \sqrt{\pi}\,\frac{(2n)!}{n!}\left(\frac{a}{2}\right)^{2n+1}$$

$$\int_0^\infty x^{2n+1} e^{-x^2/a^2}\, dx = \frac{n!}{2} a^{2n+2}$$